

Practical 1

Aim :

To design and implement a Student Course Enrollment Database using MySQL / PostgreSQL.

The objective is to create tables for Student, Course, Instructor and Enrollment, establish relationships between them, and perform database operations using SQL queries.

Theory :

- Database -

A Database is an organized collection of data that can be easily accessed, managed and updated.

- Relational Database Management System (RDBMS) -

A RDBMS such as MySQL or PostgreSQL stores data in the form of tables with relationships between them.

- In this practical, four main entities are used:

1. Student - Stores student information such as ID, name, email, age, and address.

2. Instructor - Stores instructor details such as instructor ID, name, email, and department.

3. Course - Stores course information and links each course to an instructor.

4. Enrollment - Maintains the relationship between

students and courses.

- Relationships -

- Student $\rightarrow$ Enrollment $\rightarrow$ Course

A student can enroll in multiple courses, and each course can have multiple students. This creates a many-to-many relationship, handled using the Enrollment table.

- Instructor $\rightarrow$ Course

An instructor can teach one or more courses. This represents a one-to-many relationship.

Primary keys uniquely identify records in a table while ~~foreign~~ foreign keys maintain relationships between tables.

- Keys -

- Primary key: It uniquely identifies each record in a table (eg., StudentID)

- Foreign key: It creates relationship between tables. eg., Instructor ID $\rightarrow$ Foreign key in course table

- Attributes -

It is defined as properties describing an entity.

Eg., Attributes of Student are: StudentID, Name, Email, Age, Address.

- Entity-Relationship (E-R diagram) -

The ER diagram acts as the blueprint. It illustrates how the entities interact. In this system:

- An instructor has one-to-many relationship with courses (one instructor can teach many courses).

- Student & courses have a many-to-many relationship which is resolved a functions table or link table named enrollment.

- Conclusion :

The database schema for the student enrollment system was successfully designed & implemented by using primary keys, we ensured record, uniqueness and by using foreign keys. We established logical relationships between entities, effectively mapping the ER diagram into a functional SQL database.

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ER Diagram -

Mappings
types of attributes

